

Model Selection

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Abstract

Model selection is the task of selecting a model from among various candidates on the basis of performance criterion to choose the best one. The performance of a learning method relates to its prediction capability on independent test data. Assessment of this performance is extremely important in practice, since it guides the choice of learning method or model, and gives us a measure of the quality of the ultimately chosen model. The present document is a study of two concrete techniques of model selection applied to two classification models trained on a real dataset. The study aims to elucidate if there is any significant performance difference between two given classification models trained and evaluated on the same dataset.

1 Introduction

The process of modelling starts with a given dataset D which is a subset of the population Ω . In practice, we do not have access to the population, and we must make use of the dataset to create our model. The aim of any supervised learning model is to find a function f , the target function, that assigns a value $y = f(x)$, to each $x \in \Omega$. The target function is unknown, and therefore the model is based on hypotheses about f that represent our beliefs about it. The model is then a function $\hat{f}$ that assigns a value $\hat{y}$, the output of the model, to each $x \in \Omega$. That is, it assigns a prediction $\hat{y}$ to each value $x \in \Omega$.

The performance of the model, in this context, is a measure of its capability to make predictions, i.e., how close are its predictions $\hat{y}$ to the true value y . Closeness implies a metric in the data space as a representation of the goodness of our fit, a figure of merit usually called the cost function J . Model selection is sometimes understood as an optimization problem. That is, given a family of models $\{f(y; \theta) : \theta \in \Omega_\theta\}$ depending on some parameters θ , find the model $\hat{f}(y; \hat{\theta})$ that optimize J . However, in the present document, we define the problem of model selection in different terms as a decision-making process that takes into account variability and uncertainty. Therefore, we first need to understand which are the variability and uncertainty sources in the model selection problem.

A learning algorithm is a set of instruction that tried to model the target function using a training dataset R . Since we are studying supervised learning models, the training dataset R is made of tuples $\{x, f(x)\} \in R$ where x belongs to a subset of D . A learning algorithm is therefore similar to the interpretation of model selection we saw before if we take into account the training dataset R . A learning algorithm A takes as input a training dataset R and outputs a model $\hat{f}_A$. The main problem comes from the capability of the model to make predictions for unseen data, that is, the predictions $\hat{f}(x)$ for those $x \in \Omega$ such that x is not on the training dataset.

The dataset D is usually understood as a collection of observations randomly drawn from Ω according to g_X , the probability distribution of the random variable X that generates the population¹ Ω . One possible question is how efficiently D represents Ω because it also affects our model ability to make predictions on unseen data. In general, the bigger the dataset the better it represents the population. However, several questions arise at this point, most of them referring to the relation between the learning algorithm, which include dataset properties, and the performance of our model.

As the size of the dataset D decrease, the learning algorithm becomes more sensitive to the sampling previously explained due to the variability of the data. Calculating the generalized error, a measure of the model performance on unseen data, becomes more challenging. In addition, some learning algorithms depend on random processes yielding different results even for the same training dataset.

¹We set out the difference between sample space of a random variable and population, since they are not necessarily the same.

Once we understood why model selection becomes a statistical problem, we are now in a position to introduce our model selection definition. The model selection problem is a decision-making process where we select one model between two given models $\hat{f}_A$ and $\hat{f}_B$ if one model outperforms the other with statistical significance. The statistical significance is important because it means that the differences observed in performance are unlikely to have occurred by random chance alone. The model selection problem is therefore a 2-step process: first we define a method to measure the models' performance which represents their capability to make predictions on unseen data, and then they are compared based on this measure to test if there is a difference in performance which is statistically significant.

2 Classifiers selection techniques

In this section of the document, our attention shifts towards exploring supervised learning models that are specifically designed to handle categorical outcome variables. These models are commonly referred to as classifiers. Particularly, the problem we want to address is, given two learning algorithms A and B that generate classifiers $\hat{f}_A$ and $\hat{f}_B$, respectively, and a dataset D , how to make a decision which of the two algorithms performs best based on classification accuracy J for the given dataset. The accuracy is our measure of performance, and it is defined as

$$J = \frac{1}{N} \sum_{i=1}^N \delta(L(\hat{y}_i, y_i)) \quad (1)$$

where δ is the Kronecker's delta function and L is the 0-1 loss function such that

$$L(\hat{y}_i, y) = \begin{cases} 0 & \text{if } \hat{y}_i = y_i \\ 1 & \text{if } \hat{y}_i \neq y_i \end{cases} \quad \text{and} \quad \delta(L(\hat{y}_i, y)) = \begin{cases} 1 & \text{if } \hat{y}_i = y_i \\ 0 & \text{if } \hat{y}_i \neq y_i \end{cases} \quad (2)$$

where y_i is the i -th true class label and $\hat{y}_i$ the i -th predicted class label, respectively. The accuracy of a classifier is the probability that $\hat{f}$ will classify correctly an example drawn randomly from Ω according to g_X . In practice, this accuracy is estimated from D because we do not have access to Ω . There are different methods to compute the point estimate of the accuracy and its corresponding uncertainty. Any estimator $\hat{J}$ could have a high bias², or a high variance³, both of them are undesirable, and usually we are interested in finding a good trade-off between them.

Model performance: a statistical estimate

The simplest method is the holdout method where we randomly subdivide our dataset D into two mutually exclusive subsets: a training set R and a test set T . A learning algorithm trained on R generates the model $\hat{f}$. The accuracy of $\hat{f}$ on T , denoted as J_T , provides an estimate of the true accuracy of $\hat{f}$ on the population Ω , the generalized performance of the model J_Ω . As discussed above, the point estimate $\hat{J}_\Omega$ has an associated uncertainty. A confidence interval⁴ can be computed for modelling the uncertainty if each test instance is viewed as a Bernoulli trial correct or incorrect prediction via the normal approximation.

The normal approximation relies on certain assumptions such as the statistical independence, which is actually violated in the training-test set split [7, Section 1.4]. In addition, the holdout method makes inefficient use of the data because a proportion of the data is not used for training. If the training dataset is small, the performance estimate would be pessimistically biased, assuming that the algorithm could yield a better model if it was given more data. On the other hand, if the test dataset is small the estimate is optimistically biased and has high variance. The algorithm is more likely to overfit the training set so that model fails to generalize well to data that it has not seen before, making it training-test split dependent.

This section introduces some advanced techniques for estimating the uncertainty of our estimated model performance.

²The bias is defined as $\beta = \langle \hat{J} \rangle - J$, where $\langle \hat{J} \rangle$ denotes the expected value of $\hat{J}$

³The variance is defined as $\sigma^2 = \langle (\hat{J} - \langle \hat{J} \rangle)^2 \rangle$

⁴A confidence interval is just one of the many ways to represent the uncertainty in our point estimate.

Repeated holdout One way to obtain a more robust performance estimate is to repeat the holdout method k times with different random sampling and compute the average performance over these repetitions

$$\hat{J}_\Omega = \frac{1}{k} \sum_{i=1}^k J_i \quad (3)$$

where J_i is the accuracy estimate (1) of the i -th test set T_i of size $t_i < N$. It also provides information about the model variability due to different training-test set splits. Nevertheless, the repeated holdout method suffers from independence limitations because of the overlapping between test sets in each repetition. In addition, we cannot guarantee that each sample is being tested (at least once) since some samples may never be part of the test set.

k -fold Cross-Validation The k -fold Cross-Validation is the most common technique. Cross-validation splits the data D into k approximately equal parts $D_1, \dots, D_k$, where the learning algorithm is trained on the data $D \setminus D_i$, $1 \leq i \leq k$ with one part left out at each iteration. The part D_i left out is used as the i -th test set. We compute the cross-validation performance as the arithmetic mean over the k performance estimates. The main idea is that each sample in our dataset has the opportunity of being tested. In addition, each test set is independent of the others, which is not the case for training sets. If k is small, less training data is available for the learning algorithm resulting in a possible pessimistic bias and high variance in the model. If k is high⁵, the overlapping between training sets increase resulting in high variance [5], but the estimator is approximately unbiased since most samples are available for the learning algorithm. Unfortunately, there exists no universal (valid under all distribution) unbiased estimator of variance of k -fold cross-validation as shown in Bengio and Grandvalet [1, Theorem 6]. In fact, there are some special cases such as $k = N$, also known as Leave-One-Out Cross-Validation (LOOCV). Repeated k -fold cross-validation is another valid method proposed to overcome some limitations arising during model comparison and therefore will be studied in the next subsection.

Bootstrap Another resampling method is the Bootstrap method [4]. In brief, the idea is to generate new data from a population by repeated sampling from the original dataset with replacement. The steps are similar to repeated holdout, but now we compute the accuracy of each bootstrap sample B_i and compute the model accuracy as the average over the b accuracy estimates J_i for $i = 1, \dots, b$. The bootstrap method aims to determine the statistical properties of an estimator when the underlying distribution is unknown, which is not the case. The main downside is that the training set (asymptotically) only contains a proportion of the samples from the original dataset due to the replacement. This results in a pessimistic bias for about the same reasons as k -fold cross-validation for small values of k .

Model comparison: a statistical test

The next step in our model selection definition is how to compare two given models to test if there is a difference in terms of their performance which is statistically significant. The comparison is a statistical hypothesis testing where the null hypothesis is

Null hypothesis *Let A and B represent two learning algorithms with $\hat{f}_A$ and $\hat{f}_B$ their corresponding models, respectively. The performance of both model is the same when tested on any randomly-drawn sample $x \in \Omega$ according to g_X .*

where Ω is the population and g_X the distribution of the random variable X that generates this population.

The statistical hypothesis testing are based on some assumptions about the way the data has been collected, therefore influenced by the method chosen to carry out the model performance estimate. Here we briefly introduce some recommended test given by Dietterich [2]. A further discussion is addressed in the next section when we delve into the two model techniques focus of our study.

McNemar's test [6] After applying the holdout method we end up with $|T|$ accuracy values, for each learning algorithm tested, corresponding to every sample in the test set T . Therefore, for each test value $x \in T$, we record how it was classified and construct the contingency table shown in Table 1

⁵Note that the highest possible value is $k = N$.

# of $x \in T$ misclassified by both $\hat{f}_A$ and $\hat{f}_B$	# of $x \in T$ misclassified by $\hat{f}_A$ but not $\hat{f}_B$
# of $x \in T$ misclassified by $\hat{f}_B$ but not $\hat{f}_A$	# of $x \in T$ misclassified by neither $\hat{f}_A$ nor $\hat{f}_B$

Table 1: Contingency table for hypothesis testing

where we will use the notation:

n_{00}	n_{01}
n_{10}	n_{11}

Under the null hypothesis, the two algorithm should have the same error rate, $n_{01} = n_{10}$, and (with a sufficiently large numbers n_{01} and n_{10}) the McNemar test statistic has a chi-squared distribution with 1 degree of freedom

$$\chi_1^2 = \frac{(|n_{01} - n_{10}| - 1)^2}{n_{01} + n_{10}} \quad (4)$$

where the “continuity correction” term (of -1 in the numerator) was introduced by Edwards [3] to increase its usefulness and accuracy. If the χ_1^2 is significant⁶, this provides sufficient evidence to reject the null hypothesis, in favor of the alternative hypothesis. If not, we fail to reject the null hypothesis.

Paired t test The paired t test is based on multiple accuracy samples $\{J_i\}_{i=1}^k$ obtained from repetitive tests on our data for each learning algorithm. The difference in performance between algorithms in the i -th iteration $z^{(i)}$ is computed as a simple difference between their individual performances. The quality of the test, in terms of how well it aligns with the test assumptions, is determined by the approach employed to gather these samples. The paired t test assumes that these k differences were independently drawn and follow an approximately normal distribution. Under the null hypothesis, the statistics⁷

$$t = \frac{\bar{z}\sqrt{k}}{\sqrt{\frac{1}{k-1} \sum_{i=1}^k (z^{(i)} - \bar{z})^2}} \quad (5)$$

follows a Student’s t -distribution with $k - 1$ degrees of freedom. If the t is significant, the conclusions are the same as McNemar test or any other statistical hypothesis test.

3 McNemar’s test & 5x2-fold Cross-Validated Paired t -test

In this section, we focus on two concrete model selection techniques that combine model performance estimate and model comparison because of their low probability of Type I error⁸.

5x2-fold Cross-Validated Paired t -test

The 5x2-fold cross-validation (5x2cv) paired t -test is a procedure for comparing the performance of two models that was proposed by Dietterich [2] to address shortcomings found in other methods. It involves five replications of 2-fold cross-validation. In each of the five iterations, the available data are randomly divided into two equal-sized sets S_1 and S_2 . There are two steps for each iteration: first both algorithm are trained on S_1 and their resulting models tested on S_2 (with accuracies $J_A^{(1)}$ and $J_B^{(1)}$), and then both are trained on S_2 and their resulting models tested on S_1 (with accuracies $J_A^{(2)}$ and $J_B^{(2)}$). Therefore, we end up with two estimated differences for each replication $i = 1, \dots, 5$

$$z_i^{(1)} = J_A^{(1)} - J_B^{(1)} \quad (6)$$

$$z_i^{(2)} = J_A^{(2)} - J_B^{(2)} \quad (7)$$

⁶The significance level, often denoted as α , is predetermined and represents the threshold for determining whether the results are statistically significant. If the p -value (the probability that measures how consistent the observed data is with the null hypothesis) is smaller than the significance level, then the result is considered statistically significant.

⁷The term $\bar{z}$ represents the average difference between model performances.

In this section, we have intentionally omitted the difference of two proportion because its simplicity [7, Section 4.2].

⁸A Type I error occurs when the null hypothesis is true and the method reject the null hypothesis.

where the estimated variance is

$$s_i^2 = \left[z_i^{(1)} - \bar{z}_i \right]^2 + \left[z_i^{(2)} - \bar{z}_i \right]^2; \quad \bar{z}_i = \frac{z_i^{(1)} + z_i^{(2)}}{2} \quad (8)$$

Then define the following statistic

$$\tilde{t} = \frac{z_1^{(1)}}{\sqrt{\frac{1}{5} \sum_{i=1}^5 s_i^2}} \quad (9)$$

which we call the 5x2cv $\tilde{t}$ statistic. Dietterich [2] claim that under the null hypothesis, $\tilde{t}$ has approximately a t distribution with 5 degrees of freedom.

The derivation of the $\tilde{t}$ statistic is closer to alchemy than mathematics. The method is an attempt to overcome the large t values experimentally found in k -fold cross-validated paired t -test. In particular, by means of simulations with synthetic data, Dietterich found a poor estimate of the mean difference in the performance of the two algorithm $\bar{z}$. On the other hand, the variance was slightly underestimated. The proposed statistic finds its foundation in observed favorable outcomes, particularly when alterations are introduced to the estimates such as the mean or the variance of the Student's t -distribution.

The $\tilde{t}$ is mathematically inconsistent, nevertheless it provides well-behaved results compared to other methods and here we try to justify them.

1. The required statistical independence between $z_i^{(1)}$ and $z_i^{(2)}$ for s_i/σ^2 to follow a χ_1^2 is violated. Still, the assumption of independence is probably more appropriate for 2-fold cross-validation than for 10-fold cross-validation, because the training sets are not overlapping. Obviously, it also implies that the learning algorithm are trained on smaller training sets.
2. The numerator in the $\tilde{t}$ statistic correspond to an estimate of only one replication, but the denominator is a sum of five estimates. As stated before, experimentally the variance estimate was very noisy, and often zero. The combination of five replication is shown to stabilize the variance estimate.

These arguments and experiments suggest that the 5x2cv test is the most powerful among those statistical tests that have acceptable Type I error. This test is also convenient when we want to know the effect of the choice of training set (by running the learning algorithm on five different training sets) and the choice of the test set (by measuring the performance on five test sets) [2]. Precisely these reasons make the 5x2cv test computationally more expensive. Other techniques are required to obtain acceptable results at lower cost, and the McNemar's test is shown to be a good candidate.

McNemar's test

The McNemar's test is another model selection technique with a low probability of Type I error as shown by [2]. It does not require multiple learning algorithm running, turning it into a low-cost technique in terms of computational cost. It also has some advantages over other test because it looks over absolute values of proportions and not only at their differences. This is especially notorious when the difference in error rates is not representative of the comparison, but their absolute values are. It would be also preferred over 5x2cv test in cases where the error rates measured in the various 2-fold cross-validation replications vary widely resulting in bad estimates of the variance.

The biggest drawback of the McNemar's test is that it does not assess the effect of varying the training sets. Therefore, it would be recommended only when the available data is enough to overcome this variability due to the training-test split. Obviously, the term "enough" is meaningless if we cannot quantify it, but unfortunately it begs the question because that is the whole point. We should make a decision about the most appropriate test in terms of the given problem context, knowing the limitations of each test and experimentally leading to a final decision. This is the topic of the next section where we apply McNemar's test and 5x2-fold Cross-Validated Paired t -test to a well known classification problem.

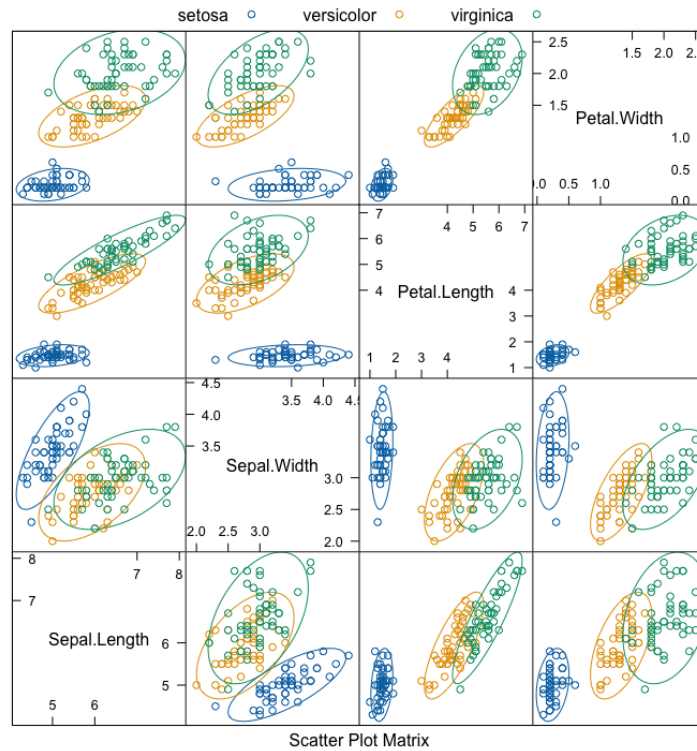
4 Results

The Iris dataset is a well-known and frequently used dataset in the field of machine learning and statistics. It was introduced by the British biologist and statistician Ronald A. Fisher in 1936 as an example of discriminant analysis. The dataset contains measurements of four features of three different species of iris flowers. For each species, there are 50 samples, resulting in a total of 150 data points. The features are

- Sepal length
- Sepal width
- Petal length
- Petal width

The three species of iris flowers in the dataset are

- Iris setosa
- Iris versicolor
- Iris virginica



This section delves into the comparative analysis of two widely employed algorithms, namely the decision ("**rpart**") tree and k-nearest neighbors ("**knn**") from the **library**(caret), using the renowned Iris dataset as the empirical foundation. These algorithms are previously evaluated for hyperparameters tuning. The models evaluations consist on a data preprocessing of centering and scaling, and a 5-fold cross-validation where accuracy was used to select the optimal hyperparameters setting using the largest value. The results are

cp	Accuracy	Kappa	k	Accuracy	Kappa
0.000	0.9400000	0.91	5	0.9333333	0.90
0.44	0.7600000	0.64	7	0.9466667	0.92
0.50	0.3333333	0.00	9	0.9466667	0.92

Table 2: Hyperparameters tuning

Therefore, the two learning algorithms to be compare are:

- A: 7-Nearest Neighbors algorithm.
- B: Decision tree with null complexity parameter⁹ (cp).

⁹The main role of this parameter is to save computing time by pruning off splits that are obviously not worthwhile

Please note how we have clearly separated hyperparameter tuning from model selection. The model selection starts once we have both learning algorithms, which include their hyperparameters' configuration. Obviously, if the dataset were bigger we can split our dataset into train, validation and split applying a simple test for model selection. Nevertheless, since we are dealing with a smaller dataset (150 samples), we decided to start the model comparison assuming that hyperparameters are already defined (fixed).

The models' performances are assessed through the McNemar's test and the 5x2cv paired t -test.

McNemar's test

To apply McNemar's test, following Dietterich [2], we divide the iris dataset into a training set R and a test set T with stratification. Training dataset represents 70 % of the dataset (105 samples). We train both algorithm¹⁰, A and B , on the training set yielding classifiers $\hat{f}_A$ and $\hat{f}_B$, respectively. Then, we record the predictions $\hat{y} = \hat{f}(x)$ of both models for each $x \in T$ and construct the contingency Table 1

$n_{00} = 0$	$n_{01} = 2$
$n_{10} = 4$	$n_{11} = 39$

Table 3: McNemar's test on iris dataset

The statistic (4) is (approximately) distributed as χ^2_1 . The observed¹¹ McNemar's chi-squared¹² is 0.16667 with a p-value 0.6831. If we apply a significant level of 0.05 we fail to reject the null hypothesis, i.e., we do not have sufficient evidence to conclude that there is a significant difference between the two compared models. Note that failing to reject the null hypothesis does not necessarily mean that the null hypothesis is true, it simply means that we have not found enough evidence to support the alternative hypothesis.

5x2-fold Cross-Validated Paired t -test

In this test¹³, we perform 5 replications of 2-fold cross validation. As explained before, the multiple training and test yield the statistic (9). The observed 5x2cv $\tilde{t}$ statistic is -0.50637 with a p-value of 0.63413. If we apply a significant level of 0.05 we fail to reject the null hypothesis. The conclusions are analogous to McNemar's test when we fail to reject the null hypothesis.

5 Conclusions

The generalization performance of predictive models cannot be directly computed because of the limited access we have to data. This performance is always an estimate of the ground-truth, endowed with its corresponding uncertainty. The uncertainty associated with our estimate depends on the amount of data available and the use we make of it. Several methods have been proposed to provide a good estimate, and the final decision always involves choosing the method that best match our problem. Therefore, the scope and limitations of each method have been described.

The model performance estimates, allow us to formulate a null hypothesis. The model comparison as a hypothesis test is based on the fact that if the null hypothesis is true, the probability distribution of the difference of model performances has zero mean. We have presented different methods to test such hypothesis, all of which are based on slightly different assumptions. All these tests assume statistical independence between accuracy samples, an assumption that is obviously violated. Nevertheless, two of them, namely McNemar's test and 5x2cv, are shown to have less probability of Type I error. Although the 5x2cv is the most powerful, the final technique could be determined by the given context of the problem such as computational cost.

¹⁰The preprocessing applied during training is the same as during evaluation.

¹¹The code used to obtain these values is `mcnemar.test(cTable)`, where `cTable` is Table 3.

¹²We also incorporate the continuity correction in our calculation.

¹³The 5x2cv paired $\tilde{t}$ -test has been applied using Python library `mlxtend` because it has a specific function for this test, namely `paired_ttest_5x2cv`. The same learning algorithm are executed, with the same hyperparameters, but using the scikit-learn implementation of them. The main difference is the preprocessing, since we have not applied it in python for this test.

The final selection will depend on the context where we are applying our models, i.e., the problem we are trying to solve. If we would like to compare learning algorithm performance, comparison may include computational cost and available test size. Indeed, if the learning algorithm involves a computationally expensive training, we should apply the McNemar’s test even though we know that it is less powerful. The main reason is the required single training instead of multiple trainings. For algorithms that can be executed multiple times, the 5x2cv test is recommended because it is not only slightly more powerful, but also because it directly measures variation due to the choice of the training set.

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